GATE-ALL BRANCHES



ENGINEERING MATHEMATICS

Probability and statistics

Lecture No.- 05



By- Rahul sir

Recap of Previous Lecture



Topic

Concepts and problems based on fundamental probability

Topics to be Covered

Something is common

Topic Problems based on events



$$P(\text{both occur}) = P(E \cap F) = \frac{n(E \cap F)}{n(S)}$$

$$P(\text{either } E \text{ or } F) = P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

$$P(E \cap F) = P(E)\,P\left(\frac{F}{E}\right) \longrightarrow \text{Dependent}$$

$$P(E \cap F) = P(E)\,P(F) \longrightarrow \text{Independent}$$

E and F independent events

$$P(\text{only } E) = P(E \cap \bar{F}) = P(E)(1 - P(F))$$

$$P(\text{only } F) = P(\bar{E} \cap F) = P(F)[1 - P(E)]$$

$$P(\text{exactly one}) = P(E) + P(F) - 2P(E)P(F)$$

- ✓ Understanding The Problem
- ✓ Devise The Plan
- ✓ execute (cheat)
- ✓ Solution



both excluded
2-199

1-200 (positive integer)
(both included)

#Q. An integer is chosen at random from 200 positive integers. Find the probability that the integer chosen is divided by 6 or 8

$$P(6 \cup 8) = P(E) + P(F) - P(E \cap F)$$

(E: 6, F: 8)

A 1/4

B 1/6

C 1/2

D N.O.T.

Using Prob. Addition Theorem

$P[E \cap F] = P[E \cup F]$

$P[E \cup F] = P(E) + P(F) - P(E \cap F)$

$= P(6) + P(8) - P(6 \cap 8)$

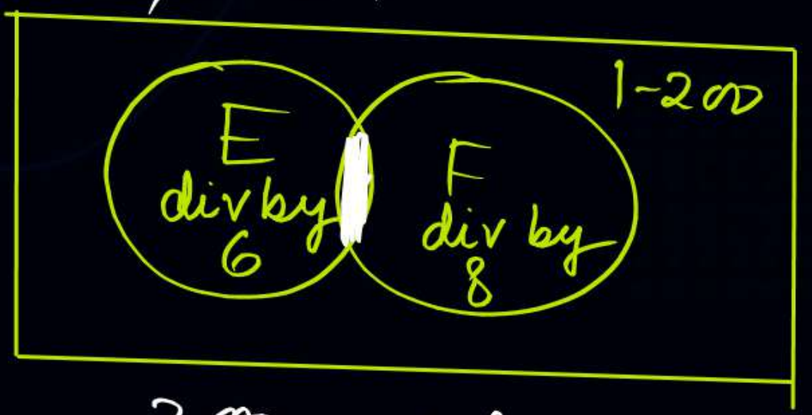
$= \frac{n(6)}{200} + \frac{n(8)}{200} - \frac{n(6 \cap 8)}{200}$

$= \frac{33 + 25 - 8}{200}$

$= \frac{50}{200}$

$\boxed{P(6 \cap 8) = \frac{1}{4}}$

$8, 6 = 24$



$\frac{200}{6}$ Total No = 33

$\frac{200}{8}$ Total No = 25

$\frac{200}{24} = 8$

(24) ~~24~~



#Q. Two cards are drawn from pack of 52 cards. What is the probability that either of the cards, both are red or both are kings

A 660/2652

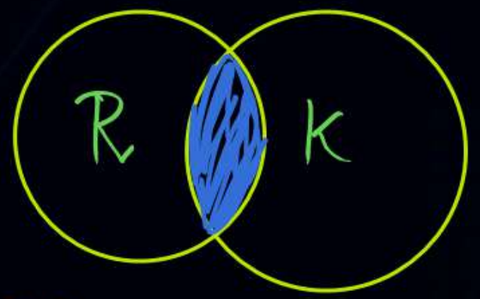
B 660/1352

C 330/2652

D 1/52×53

26 cards: red
King – 4
52 cards
2 card (R+K)

→ Red cards
→ King cards



Red card = 26
King card = 4

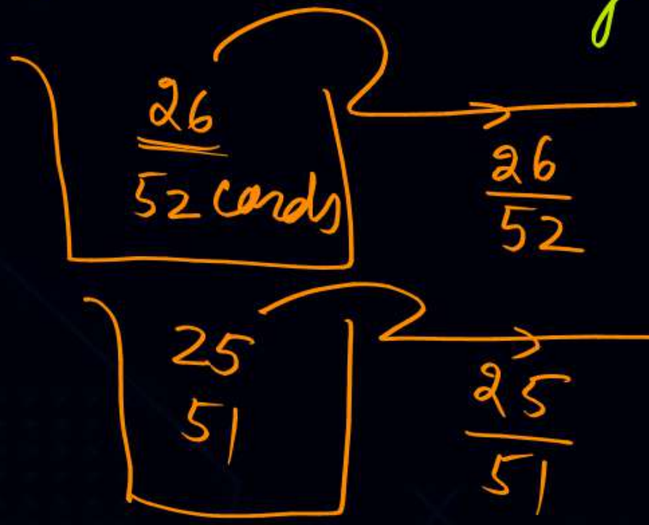
$P(\text{red or king}) = P(R \cup K)$

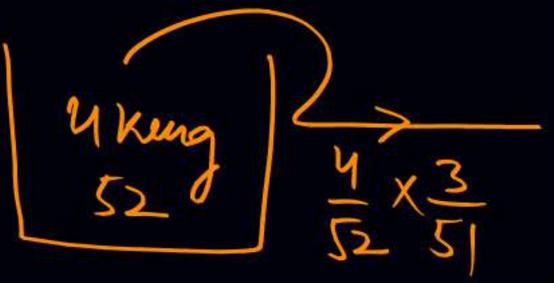
$P(R \cup K) = P(R) + P(K) - P(R \cap K)$

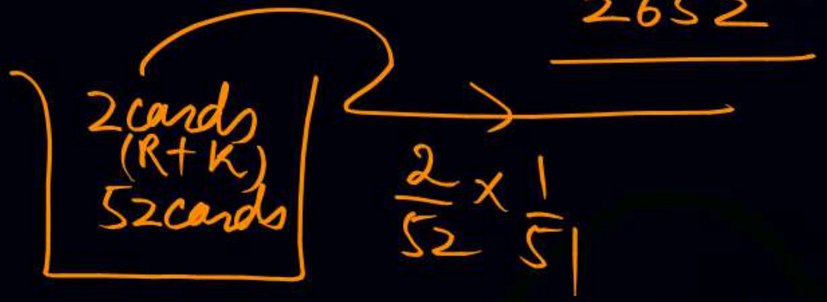
$= \frac{26 \times 25}{52 \times 51} + \frac{4 \times 3}{52 \times 51} - \frac{2}{52} \times \frac{1}{51}$

$= \frac{660}{2652}$

2 [HEART red and king
Diamond red and king









$P(A)=\frac{3}{4}$ $P(B)=\frac{1}{2}$ $P(C)=\frac{5}{8}$

A, B, C (Independent)

#Q. 3 person A, B, C independently try to hit a target, if probability of hitting a target by A, B, C are 3/4, 1/2, 5/8 then the probability that target hit by A or B but not C

A 21/64

B 1/64

C 7/64

D 3/64

$P(A)=\frac{3}{4}$ $P(B)=\frac{1}{2}$ $P(C)=\frac{5}{8}$ → something is common

$P[(A \cup B) \cap \bar{C}] \Rightarrow P(A \cup B) \cdot P(\bar{C})$

$\Rightarrow [P(A)+P(B)-P(A \cap B)]\, P(\bar{C})$

$\Rightarrow \left[\frac{3}{4}+\frac{1}{2}-P(A)P(B)\right]\left[1-P(C)\right]$

$\Rightarrow \left[\frac{3}{4}+\frac{1}{2}-\frac{3}{4}\times\frac{1}{2}\right]\left[1-\frac{5}{8}\right]$

$\Rightarrow \frac{21}{64}$ Ans

A OR B but Not C

$(A \cup B)$ but Not C

system 1 ∩ system 2

$=[(A \cup B) \cap \bar{C}]$

Happen $\begin{bmatrix} A \\ B \\ C \end{bmatrix}$ Independent

Does Not Happen $\begin{bmatrix} \bar{A} \\ \bar{B} \\ \bar{C} \end{bmatrix}$ also Independent

$P(A \cap B)=P(A)\,P(B)$



E and F independent events

$P[E \text{ and } F] = \frac{1}{12}$ $P(E \wedge F) = \frac{1}{12}$ — (1)

#Q. Let E and F be two independent events. The probability of both E and F happens is 1/12 and neither E nor F happens is 1/2 then the value $\frac{P(E)}{P(F)} = ?$ $\frac{P(E)}{P(F)}$

$P[\bar{E} \wedge \bar{F}] = \frac{1}{2}$ — (2)

Something is common

A 4/5

B 3/2

C 1/5

D N.O.T.

$P[E \wedge F] = \frac{1}{12}$ $\quad P(\bar{E} \wedge \bar{F}) = \frac{1}{2}$

'E and F are independent events $\quad P(E \wedge F) = P(E)P(F)$

$P(\bar{E} \wedge \bar{F}) = P(\bar{E})\,P(\bar{F})$

$P(E)\,P(F) = \frac{1}{12}$ $\quad P(\bar{E})\,P(\bar{F}) = \frac{1}{2}$

$$x \cdot y = \frac{1}{12} \quad \text{—— (1)}$$

$$(1-P(E))(1-P(F)) = \frac{1}{2}$$

$$(1-x)(1-y) = \frac{1}{2}$$

$$1 - y - x + xy = \frac{1}{2}$$

$$1 + \frac{1}{12} - \frac{1}{2} = x + y$$

$$x + y = \frac{1}{2} + \frac{1}{12} = \frac{14}{24} = \frac{7}{12}$$

(2) —— $x + y = \frac{7}{12}$

If something is common

$P(E) = x$

$P(F) = y$

Nothing is common

$P(E) = x$

$P(F) = 1 - x$

$x \cdot y = \frac{1}{12} \qquad x + y = \frac{7}{12}$

$(x-y) = \sqrt{(x+y)^2 - 4xy}$

$= \sqrt{\frac{49}{144} - \frac{4}{12}}$

$(x-y) = \sqrt{\frac{49-48}{144}} = \pm\frac{1}{12}$

$x+y = \frac{7}{12}$

$xy = -\frac{1}{12}$

$x+y = \frac{7}{12}$

$x-y = \frac{1}{12}$

$2x = \frac{8}{12}$

$x = \frac{8}{24} = \frac{1}{3}$

$\frac{1}{3} + y = \frac{7}{12}$

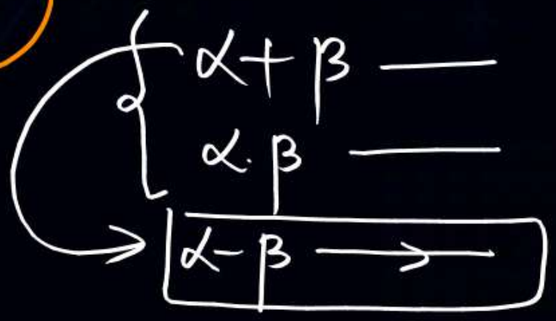
$y = \frac{7}{12} - \frac{1}{3}$

$= \frac{21-12}{36} = \frac{9}{36} = \frac{1}{4}$

$y = \frac{1}{4}$

$\frac{P(E)}{P(F)} = \frac{\frac{1}{3}}{\frac{1}{4}} = \frac{4}{3}$







$P(E \cup F) = P(E) + P(F) - P(E \cap F)$

$= P(E) + P(F) = 1$

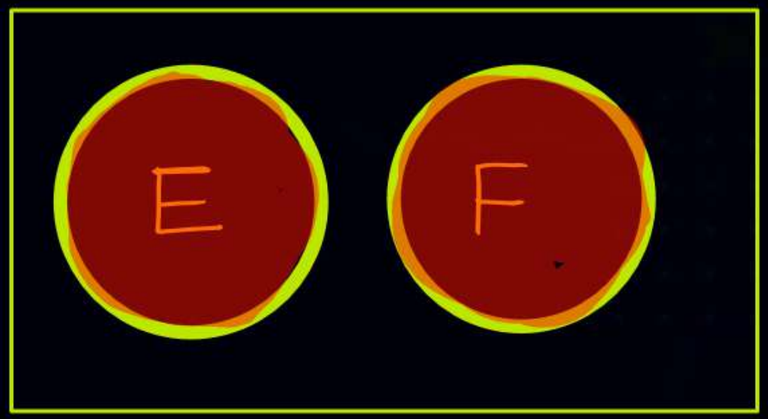
$P(F) = 1 - P(E)$

Mutually exclusive events :

If E and F Are mutually exclusive (Nothing is Common)

E and F both are Disjoint events $P(E \cap F) \overset{U}{=} 0$

$$P(E \cap F) = 0 \text{ or } \phi$$



Statistically E and F (Happening) → same time Not Possible

- Win / Loose ✓ ✓
- H / T
- 6 / $\bar{6}$
- Late / on time

$$P(E \text{ or } F) = P(E) + P(F) - P(E \cap F)$$

$$P(E \cup F) = P(E) + P(F) - 0$$

If E and F both are mutually events

$$P(E \cup F) = P(E) + P(F)$$

$$P(E \cup F \cup G) = P(E) + P(F) + P(G)$$

E, F, G Three of Them are mutually exclusive events





$P(S) = 1$

#Q. Let S be the sample space with two mutually exclusive events A and B and $A \cup B = S$. (sure event) If P denotes probability of events, then the maximum value of $P(A) \cdot P(B) = ?$

A 1/2

B 1/4

C 1/6

D N.O.T.

$P(A \cup B) = P(S)$ → sure event = 1

$\Rightarrow \quad P(A) + P(B) = 1$

$P(B) = 1 - P(A)$

Max value $= P(A)\,P(B) = P(A)\,[1 - P(A)]$

$(P(A)\,P(B))$ $\quad = \quad x(1-x)$

$f(x)$

$f(x) = x(1-x)$

$f'(x) = 1 - 2x$

$f'(x) = 0$

$1 - 2x = 0$

$\boxed{x = \frac{1}{2}}$

$\boxed{x = \frac{1}{2}}$

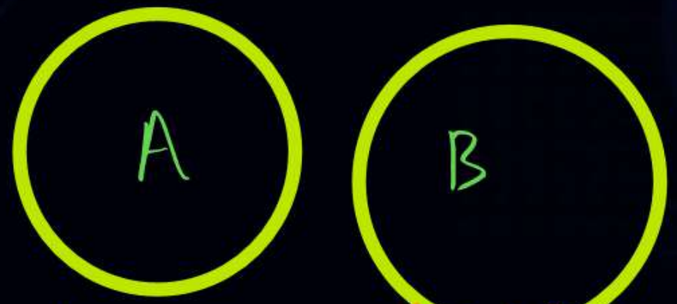
$f''(x) = 0 - 2$

$f''(x) < 0$ (maxima at Point $x = \frac{1}{2}$)

max value $f(x) = x(1-x)$

$= \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{4}$

$\boxed{\text{Max value} = \frac{1}{4}}$



A and B mutually

$P(A) = x$

$P(B) = (1-x)$

A.M. G.M inequality

$$A.M \geq G.M$$

$$= \frac{a+b}{2} \geq \sqrt{ab}$$

$$\frac{P(A)+P(B)}{2} \geq \sqrt{P(A)\,P(B)}$$

$$= \left(\frac{1}{2}\right)^2 \geq \left[P(A)\,P(B)\right]_{max}$$

$$= \frac{1}{4} \geq \left(P(A)\,P(B)\right)_{max}$$



MSQ

P(exactly one of them) = $\frac{11}{25}$ P(None of them) = $\frac{2}{25}$

#Q. Let E and F be two independent events. The probability that exactly one of them occurs is 11/25 and the probability of none of them occurring is 2/25. If P(T) denotes the probability of occurrence of the event T, then

A P(E) = 4/5, P(F) = 3/5

B P(E) = 1/5, P(F) = 2/5

C P(E) = 2/5, P(F) = 1/5

D P(E) = 3/5, P(F) = 4/5

P[exactly one of them]

$$= P(E \cap \bar{F}) + P(\bar{E} \cap F)$$
$$= P(E)P(\bar{F}) + P(\bar{E})P(F)$$
$$= P(E) + P(F) - 2P(E)P(F)$$

$$\begin{cases} \frac{11}{25} = P(E) + P(F) - 2P(E)P(F) \\ \frac{2}{25} = [1-P(E)][1-P(F)] \end{cases}$$

Ram Job

Shayam

$(R \cap \bar{S}) + (S \cap \bar{R})$

Individual

$x + y - 2xy = \frac{11}{25}$ —①

$(1-x)(1-y) = \frac{2}{25}$ —② quadratic

$P(E) = x$
$P(F) = y$

$P(E) = \frac{4}{5}$ $P(F) = \frac{3}{5}$

$P(E) = \frac{3}{5}$ $P(F) = \frac{4}{5}$ ✓

(A) (D)



#Q. If $\bar{E}$ and $\bar{F}$ are the complementary events of E and F respectively and if $0 < P(F) < 1$. then;

A $P(E/F) + P(\bar{E}/F) = 1$

B $P(E/F) + P(E/\bar{F}) = 1$

C $P(\bar{E}/F) + P(E/\bar{F}) = 1$

D $P(\bar{E}/F) + P(\bar{E}/\bar{F}) = 1$



#Q. Let E and F be two independent events. If the probability that both E and F happens is 1/12 and the probability that neither E nor F happens is 1/2. Then

A $P(E) = 1/3, P(F) = 1/4$

B $P(E) = 1/2, P(F) = 1/6$

C $P(E) = 1/6, P(F) = 1/2$

D $P(E) = 1/4, P(F) = 1/3$



#Q. The probability that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the students has a 75% chance of passing in atleast one, a 50% chance of passing in atleast two and a 40% chance of passing in exactly two. Which of the following relations is true?

A $p+m+c = 19/20$

B $p+m+c = 27/20$

C $pmc = 1/10$

D $Pmc = 1/4$

THANK - YOU